Neuroscience and Biobehavioral Reviews 159 (2024) 105572



Contents lists available at ScienceDirect

Neuroscience and Biobehavioral Reviews

journal homepage: www.elsevier.com/locate/neubiorev





Effects of internet and smartphone addiction on cognitive control in adolescents and young adults: A systematic review of fMRI studies☆

M. León Méndez [a,1], I. Padrón [a,c,*,2], A. Fumero [a,b,3,4,5], R.J. Marrero [a,b,6,7,8]

[a] Instituto Universitario de Neurociencia, Universidad de La Laguna, La Laguna, Santa Cruz de Tenerife 38200, Spain
[b] Department of Clinical Psychology, Psychobiology, and Methodology, Spain
[c] Department of Developmental Psychology and Education, Spain

ARTICLE INFO

Keywords:
Internet addiction
Smartphone addiction
Executive functions
FMRI
Young adults
Adolescents

ABSTRACT

Background: Internet Addiction (IA) refers to excessive or uncontrolled behaviors related to Internet access, causing impairment or distress. Similarly, Smartphone Addiction (SPA) involves the excessive use of smartphones, disrupting users' daily lives. Cognitive control plays a relevant role in the development of IA and SPA, with executive functions and rewards processing being particularly important.
Objective: This study aims to provide a synthesis of scientific evidence on the differential effects of IA and SPA on cognition in young adults and adolescents, using fMRI.
Methods: Relevant articles published between 2013 and May 2023 were identified following the PRISMA protocol.
Results: Included studies (n = 21) revealed that both behavioral addictions were associated with impairments in cognitive control related to reward processing (ACC, insula, amygdala) and executive function (DLPFC, frontal, parietal lobes) in adolescents and young adults.
Conclusion: These findings provide evidence for neuroanatomical similarities between IA and SPA in young adults and adolescents related to executive functions and rewards processing. However, differential effects of SPA and IA on cognitive control were not found in this study. Future research should explore the relationship between these addictions with other cognitive domains to further expand our understanding within this field. While both IA and SPA involve the excessive use of online technology, SPA could involve a higher risk, given that smartphones allow internet connectivity anytime.

1. Introduction

Internet addiction (IA) and smartphone addiction (SPA) are increasing worldwide, particularly among adolescents and young adults (Ballestar-Tarín et al., 2020; Kwon et al., 2013; Meng et al., 2022; Olson et al., 2022), leading to significant changes in people's lives, in many cases affecting their biopsychosocial health (Cai et al., 2023; Kwak et al., 2022). These behaviors encompass a range of activities including information seeking, communication, shopping, entertainment, gaming, gambling and sex. IA is defined as the excessive or poorly controlled

☆ PROSPERO preregistration: CRD42023437105

* Correspondence to: Departamento Psicología Evolutiva y de la Educación, Campus de Guajara, Universidad de La Laguna, Apartado 456, La Laguna, CP 38200 Santa Cruz de Tenerife, Spain.
E-mail addresses: mleonmen@ull.edu.es (M. León Méndez), ivpadron@ull.edu.es (I. Padrón), afumero@ull.edu.es (A. Fumero), rmarrero@ull.edu.es (R.J. Marrero).
[1] 0009-0004-2783-6120
[2] 0000-0001-8093-5225
[3] 0000-0001-7057-6644
[4] WoS Identifier: G-8597-2011
[5] SCOPUS Identifier: 15051426500
[6] 0000-0002-3068-0196
[7] WoS Identifier: D-2259-2014
[8] SCOPUS Identifier: 6701463024

https://doi.org/10.1016/j.neubiorev.2024.105572
Received 31 August 2023; Received in revised form 1 February 2024; Accepted 2 February 2024
Available online 5 February 2024
0149-7634/© 2024 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

preoccupations, urges, or behaviors associated with computer and Internet use that result in impairment or distress (Shaw and Black, 2008). IA have been linked to anxiety disorders, depression, attention deficit and hyperactivity disorders, stress, low self-esteem, social anxiety, and poor psychological well-being (Choi et al., 2015; Kwak et al., 2022; Ponnusamy et al., 2020; Salama, 2020). Similarly, SPA has been described as the excessive use of smartphones that disrupts the user's daily life (Demirci et al., 2014). Symptoms such as preoccupation with the device, loss of control over use, socially dangerous use, feelings of anger or tension when the phone is inaccessible, ringer anxiety, constant worry about battery drain, craving, need for constant upgrades, and poor performance with social isolation may indicate SPA (Nikhita et al., 2015). Thus, both IA and SPA, like substance-related addictions (Kuss and Griffiths, 2017), have been shown to produce tolerance, salience, withdrawal and craving (Kuss et al., 2014; Kwon et al., 2013; Van Rooij and Prause, 2014).

Although IA and SPA have in common the use of the Internet for different activities, some researchers emphasize the importance of distinguishing between them due to potential differences in their impact on the mechanisms of developing and maintaining behavioral addiction, as well as different patterns of addiction risk factors (Jeong et al., 2019; Montag et al., 2021). Furthermore, these behaviors can be distinguished from specific online activities such as chatting, gaming or shopping, among others (Pontes and Griffiths, 2014).

However, the debate continues as to whether this excessive use of technology qualifies as an addictive behavior or meets the criteria for recognition as a disorder under the DSM-5 (American Psychiatric Association, 2013), as it is unclear whether it causes functional damage at a physiological level or is maintained over time (Billieux et al., 2015; Pies, 2009).

Neuroimaging techniques, particularly fMRI, provide a valuable tool for improving our understanding of the neural correlates of different addictive disorders, highlighting both similar and different underlying neural abnormalities in behavioral addictions. To study the effects of IA and SPA on cognition, fMRI's ability to capture detailed brain images in real time is essential. It allows researchers to observe neural activity associated with cognitive processes and examine changes in neural networks associated with attention, decision-making and impulse control - critical components of cognitive function - by analyzing blood flow and changes in oxygen levels in different regions of the brain. This method provides a comprehensive understanding of how addictive behaviors affect the brain at a neurological level.

Cognitive control mechanisms refer to those processes that allow the individual to encode information and select the most appropriate response as well as to inhibit less appropriate ones to environmental requirements, through the various cognitive processes involved in information processing, attention, working memory, organized search, internal models, flexibility of thought and action (Logan, 1988; Luria, 1973). Several studies have been conducted on the neuroanatomical effects associated with IA and SPA (Alavi et al., 2012; Kim et al., 2017). There is strong empirical evidence about the relationship between cognitive control and the reward system/executive control system, specifically functional alterations in three brain regions are particularly important in the onset, development and maintenance of addiction: the basal ganglia, the extended amygdala and the prefrontal cortex. These changes can (i) increase incentive salience; (ii) decrease the sensitivity of brain systems involved in the experience of pleasure or reward; and (iii) reduce the functioning of executive control brain systems involved in the ability to make decisions and regulate one's actions, emotions, and impulses (US Department of Health and Human Services, 2016). According to the I-PACE model, the individual's predisposition, the cognitive processes (i.e. attentional biases), the affective processes (i.e. mood or emotional regulation), and the executive functions (i.e. inhibitory control), would explain decision-making behavior when using the Internet (Brand et al., 2016). Situational and personal factors interact, leading to changes in mood that cause individuals to focus attention on short-term rewards and make risky decisions. Thus, Internet addiction has been found to be associated with alterations in the prefrontal cortex related to a greater salience of stimuli and a decreased inhibitory control that affects decision-making (Brand et al., 2019). Both executive and affective functions affect cognitive control ability, which is reflected in an imbalance in the fronto-striatal circuit associated with Internet addiction (Brand et al., 2019).

Problematic use of technology related to smartphones and the internet has been consistently associated with poor decision making related to disrupted reward processing and a preference for immediate rewards over long-term gains (Alavi et al., 2012). Reward is a natural process in which the brain associates various stimuli (substances, situations, events or activities) with a positive or desirable outcome. Some studies have found impairments in regions involved in the reward system, such as reduced neural activity in the anterior cingulate cortex (ACC) during reward processing in people with problematic smartphone use (Horvath et al., 2020). In addition, an increase in grey matter volume in the bilateral putamen and right nucleus accumbens and a decrease in grey matter volume in the orbitofrontal cortex (OFC) have also been reported in people with problematic internet use (Altbäcker et al., 2016). The OFC, as part of the prefrontal cortex, is involved in executive function, decision-making and the regulation of impulsivity. Abnormalities in the OFC may interfere with action plans of people with addiction, leading them to prefer immediate rewards to delayed gratification (Volkow et al., 2013). This is because the OFC is connected to some striatal regions, including the caudate and nucleus accumbens, which associate it with reward function (Schoenbaum et al., 2006).

In addition, another neuroanatomical effect related to behavioral addictions that has also been studied is executive functions. Executive functions, or cognitive control functions, can regulate and coordinate a wide variety of cognitive processes in the performance of complex cognitive tasks. The fundamental purpose of cognitive control is to produce coordinated, orderly, and purposeful behavior in a flexible and optimized way (Crone and Steinbeis, 2017). Researchers generally agree that executive functions involve three basic subfunctions: switching between tasks or mental sets (cognitive flexibility); updating and monitoring working memory representations (working memory); and inhibiting impulsive responses (inhibitory control) (Jarmolowicz et al., 2013).

From a cognitive neuroscience perspective, executive functions may be adversely affected by IA and SPA. Such addictive behaviors may induce permanent changes in the neural network and impair the functioning of the prefrontal cortex, particularly in terms of top-down control. Executive dysfunction is increasingly observed in people with addictive behaviors (Bates et al., 2006; Rosselli and Christopher, 2019). Several studies have shown that people with IA exhibit executive dysfunctions including impulsive behaviors (Cao et al., 2007; Salehi et al., 2023; Ünsalver and Aktepe, 2017), risky decision making (Seok et al., 2015) and working memory impairments (Shafiee-Kandjani et al., 2019), among others. These functions are strongly associated with parts of the prefrontal cortex, particularly the dorsolateral prefrontal cortex (DLPFC) (Bari and Robbins, 2013; Yuan and Raz, 2014). The DLPFC is a region of the frontal lobes most commonly associated with higher order processes such as decision-making, reasoning, impulse control, inhibition and executive functions including working memory and selective attention (Curtis, D'Esposito, 2003).

Some neuroimaging studies have found impairments in inhibition and deficits in self-regulation in people with IA (Cabañas, Korzeniowski, 2015; Gao et al., 2019; Wu et al., 2013). However, not only inhibitory control was found to be impaired. According to Ioannidis et al. (2019), problematic internet use was associated with impairments in global executive functions such as inhibitory control, decision-making and working memory.

Although some studies have examined the impact of IA and SPA, together with other behavioral addictions, on reduced cognitive control (Li et al., 2015), increased reward-seeking behavior (Yoon et al., 2015)

and impaired psychological health (Billieux et al., 2015; Marciano et al., 2020; Younes et al., 2016), the results appear sparse and somewhat confusing, combining several generalized behavioral addictions, such as IA and SPA, with specific addictions, such as gaming or social networking. Therefore, the aim of this systematic review is to provide a general synthesis of fMRI studies on differential adverse effects on cognition in young adults and adolescents by identifying similarities and differences between IA and SPA. Focusing on these general addictions, rather than specific behavioral addictions, is crucial as a higher proportion of young people use the Internet for a range of activities, while a smaller percentage are involved in specific addictions. Thus, understanding the neuroanatomical effects of continued internet use could raise awareness of the impact of these behaviors on young people's development and mental health.

2. Methods

2.1. Study eligibility

This review included (P) studies of adolescents and young adults aged 13–25 years with IA or SPA. The age range of 13 to 25 years was chosen due to its association with the period prone to the development of IA (Tossell et al., 2015). This choice is based on existing literature that identifies this interval as critical for the formation of habits and behaviors related to technology, including IA and SPA. (I) This study used fMRI to investigate how IA/SPA affects cognition. The use of functional magnetic resonance imaging (fMRI) as an evaluation method is justified due to its ability to provide a detailed insight into the neural correlates associated with IA and SPA. This method was chosen because it offers a comprehensive analysis of cognitive changes, with a specific emphasis on cognitive control and the reward system. In this case, (C) studies were selected that compare individuals with IA and SPA with controls (without IA and/or SPA) to detect dysfunctions in brain regions related to executive functions and reward system. This comparison is crucial for identifying specific patterns of brain activity associated with IA and SPA. (O) The aim of these studies was assessing differences in cognition (specifically cognitive control and reward system), focusing (S) on randomized clinical trials and quasi-randomized trials study designs. We have included randomized clinical trials and quasi-randomized trials published between 2013 and 2023. Included articles had to be published in English or Spanish in peer-reviewed journals.

Reviews, meta-analyses, observational studies, theses, opinion articles and monographs have been excluded. To be eligible for inclusion, empirical studies must collect data from fMRI, specifically results of neural correlates in IA and/or SPA. Any study that does not present measures of IA and SPA through fMRI will be excluded. The decision to exclude other forms of technology addiction was made to allow for a more in-depth analysis of the selected topics, as the inclusion of a wider range of technological addictions could lead to a more superficial treatment of each addiction and make it difficult to draw clear conclusions about the neural correlates and cognitive effects associated with each addiction. In addition, IA and SPA have been widely studied and recognised as important impairments, with a substantial body of literature exploring their impact on mental health and cognition (Daniyal et al., 2022; Ioannidis et al., 2019; Seki et al., 2019). The prevalence of addiction to other specific technologies may differ, and there may be limited scientific literature establishing their distinct neural correlates and cognitive effects (Cheng et al., 2021; Dieter et al., 2017; Wittek et al., 2016).

2.2. Electronic searches and data extraction

Following the PRISMA guidelines (Page et al., 2021), articles were identified through the PubMed, ScienceDirect and Web of Science databases from 2013 to May 2023. Database searches have been saved and alerts have been used to notify authors of the publication of new articles that match the searches. The electronic search had been complemented by hand searching of references of included articles and published in previous reviews. The key terms used in databases were technology addiction (or internet addiction disorder or smartphone addiction or problematic phone use) NOT (gaming disorder or gaming addiction or video games) AND neuroimaging (or functional magnetic resonance imaging or MRI functional neuroimaging or neurological diagnostic techniques) AND young adult (or adult or adolescent). Other search restrictions were the age of the sample ("13–18" AND "19–25″), and the language of publication ("*English; Spanish*").

The initial search was conducted by the first author (ML) in the databases and all duplicate articles were removed using the Mendeley Desktop program. Studies that met the eligibility criteria were then identified based on title and abstract. The full texts of the selected articles were then analyzed by two evaluators to confirm that they met the inclusion criteria. When the first author (ML) had doubts about any article, consensus was reached by discussion with the second author of this review (IP). The agreement between reviewers for the inclusion and exclusion of the studies was unanimous. Following this process, the first author (ML) extracted the data from the included articles and analyzed it to determine the quality and risk of bias in the studies, and this appraisal process was then reviewed by the third (RM) and fourth author (AF).

2.3. Analysis

To analyze and assess the methodological quality STROBE checklist (Vandenbroucke et al., 2007) was used for reports of case-control studies, with a detailed explanation and justification for each of the 22 proposed items. Each item was rated as 0 (not satisfying requirements), 1 (satisfying requirements partially) and 2 (satisfying requirements). These items are structured into 6 sections: title and abstract, introduction, method, results, discussion and other information.

3. Results

3.1. Details of included studies

After searching selected databases for this review, a total of 3155 articles were found (258 from PubMed, 255 from Web of Science and 2642 from Science Direct). Six additional records were identified from other sources. All duplicate articles (263) were removed. This left a total of 2898 studies. Based on the title or abstract, 2831 articles were excluded, leaving a total of 67 articles to be assessed. After analyzing the full text of the selected articles to determine whether they met the criteria, 46 articles were removed. These studies were excluded for the following reasons: The sample was not within the age range of 13–25 years (3) or included participants with another technology dependence (17); they were bibliographic or opinion articles (4); or they used another neuroimaging technique (13). Finally, 21 studies were included. Fig. 1 illustrates our applied literature search using a PRISMA flowchart..

3.2. Methodological quality

The STROBE checklist (Vandenbroucke et al., 2007) was used to analyze and score the methodological quality. The evaluated studies scored between 36 and 41 points, with an average of 37.76 points. Seven articles scored 38 points (33.33%), five articles scored 37 points (23.81%), four articles (19.04%) scored 36 points, three articles (14.28%) scored 39 points, one article (4.76%) scored 40 points and one article (4.76%) scored 41 points (see Supplementary Material). Despite the relative heterogeneity of these studies, there was consistency across studies in terms of quality criteria.

According to the results of all studies in this review, 66.67% found differences in the functional connectivity of several frontal regions, such

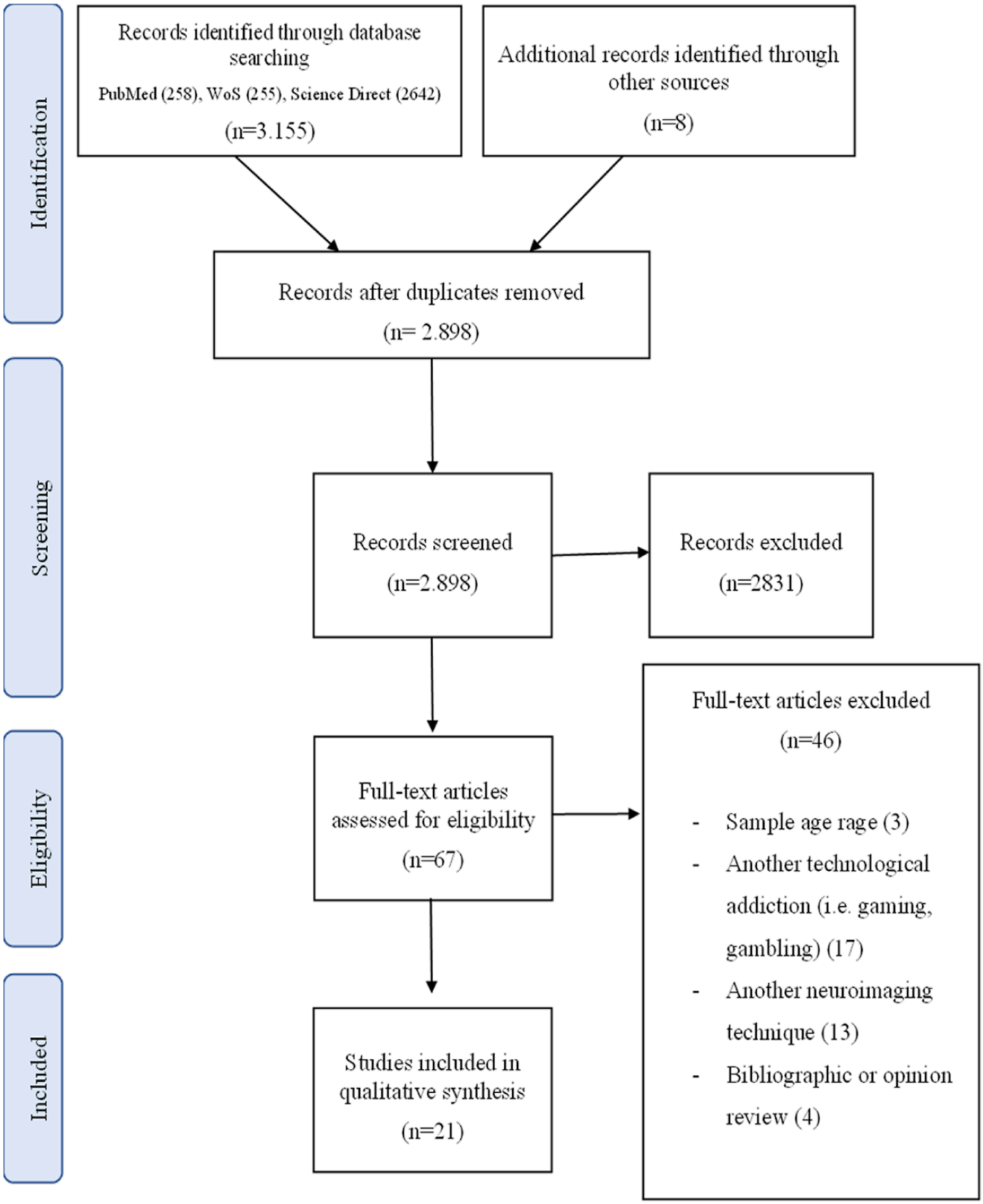


Fig. 1. Flow diagram of systematic review.

Table 1
Resting- state condition in adolescents.

ARTICLE	SAMPLE (M;F)	IA Scale	BRAIN AREA	RESULTS
Siste et al., 2022	IA: 28 (14;14) (M=14,4; SD=1,6) GC: 29 (11;18) (M=13,9; SD=1,5)	IAT	DMN	Greater FC between the left lateral PFC and the left anterior insula and decreased connectivity between the left lateral PFC and the right mPFC and lateral parietal (DMN)
Wang et al., 2017	IA: 31 (21;10) (M=15,0; SD=1,3) GC: 50 (35;15) (M= 15,1; SD=0,5)	YDQ	Fronto-parietal and DMN	FC reduction in right fronto-parietal circuit, mPFC and anterior DMN, FC reduction between salience and anterior DMN and increase FC left FPN functional connectivity.
Wee et al. (2014)	IA: 17 (15;2) (M=17,3; SD=2,6) CG: 16 (14;2) (M=17,7; SD=2,5)	YDQ	Whole brain	Stronger FC between the left fusiform gyrus and right angular gyrus, and between left angular gyrus and right middle OFC. Decreased FC between the right caudate and right supramarginal gyrus.
Hong et al. (2013)	IA: 12 (12;-) (M=13,41; SD=2,31) GC: 11(11;-) (M=14,81; SD=0,87)	IAS	Whole brain	Low FC between cortico-subcortical circuits and prefrontal, and between cortico-subcortical circuits and parietal. Bilateral putamen most involved.

Note: IAT: Internet Addiction Test; YDQ: Young's Diagnostic Questionnaire; IAS: Internet Addiction Scale.

as the dorsolateral prefrontal cortex (19.05%) and the medial prefrontal cortex (14.29%), among others. Functional impairments were also found in the limbic system (52.38%), with the anterior cingulate cortex (23.81%) being one of the most important regions reported. Changes in the other lobes were also reported. 47.62% of the articles in this review reported impairments in the parietal lobe, 23.81% in the insular cortex and a further 23.81% in the occipital regions. Impairments were also found in the temporal regions (14.29%) and 2 studies (9.52%) found impairments in the cerebellar system.

3.3. Resting-state condition

3.3.1. Adolescents

Four studies using resting-state fMRI to investigate the functional abnormalities in adolescents with IA were found. The search did not identify any studies in adolescents with SPA.

Studies in this section have found decreased functional connectivity (FC) in cortico-subcortical circuits (prefrontal and parietal) (Hong et al., 2013). Specifically, Wee et al. (2014) found decreased FC between the right caudate and right supramarginal gyrus in adolescents with high IA scores compared to controls. Furthermore, adolescents with high IA scores were found to have reduced FC in the right frontoparietal circuit, medial prefrontal cortex (mPFC) and anterior default mode network (DMN) compared to controls (Siste et al., 2022; Wee et al., 2014). In adolescents with IA, reduced functional connectivity was found between salience and the anterior DMN, and increased connectivity in the left frontoparietal network (FPN) (Wang et al., 2017), and between the left lateral prefrontal cortex (PFC) and the left anterior insula (Siste et al., 2022). Higher FC was found between the left fusiform gyrus and the right angular gyrus, and between the left angular gyrus and the right medial OFC (Wee et al., 2014). Moreover, a reduction in the FC of the

Table 2
Resting- state condition in young adults.

ARTICLE	SAMPLE (M; F)	IA Scale	BRAIN AREA	RESULTS
Afrá et al., (2023)	IA: 72 (36; 36) (M= 24.69; SD=3.23)	PIU-Q BSMAS SABAS	Whole brain	Negative correlation between FC in the medial visual network and the obsession subscale of PIU-Q. Decreased FC between the occipital pole network and auditory network in young adults with higher scores on PIU-Q obsession subscale and higher social addiction scores.
Lee et al. (2021)	IA: 39 (29;10) (M=22,9; SD= 2,2) GC: 49 (32;17) (M=22,4; SD=2,7)	IAT SAPS	Attentional networks	Dorsal:stronger FC right intraparietal sulcus - right middle occipital gyrus. Weaker FC right frontal – right DLPFC. Ventral: no differences found in FC.
Patil et al. (2021)	IA: 28 (10;18) (M=23,1;1,92) AI2: 34 (10;24) (M=63,0; SD=7,65)	CIAS-R	Prefrontal and cerebellum	Interruption of the executive-cerebellar circuit. Increased occipital-putamen connectivity
Cheng and Liu (2020)	IA: 24 (18;6) (M=20,7; SD=1,9) GC: 28 (22;6) (M=21,0 SD=1,6)	IAT	Amygdala	Negative FC: Decreased amygdala-DLPFC, Increased amygdala-precuneus-. SOG Positive FC: Decreased amygdala-ACC Increased amygdala-thalamus. Significant correlation left amygdala - right DLPFC with IA duration. Decreased connectivity and integrity amygdala – ACC.
Wang et al. (2019)	IA: 28 (21;7) (M=21,32; SD=1,96) CG: 30 (22;8) (M=21,73; SD=2,08)	IAT	Cognitive Control Network, Default Mode Network, and Visual Attention Network	Greater FC in right DLPFC, left parahippocampal gyrus, cerebellum, and the bilateral middle cingulate cortex, and superior temporal pole. Decreased FC in the right inferior parietal lobe,bilateral calcarine and lingual gyrus in IA. Significant correlations between IAT score and altered FC values in left parahippocampal gyrus and bilateral superior temporal pole.

(*continued on next page*)

Table 2 (*continued*)

ARTICLE	SAMPLE (M; F)	IA Scale	BRAIN AREA	RESULTS
Li et al. (2015)	IA: 260 (120;140) (M=19,9; SD=1,2)	IAT	Whole brain	Low anticorrelation between IAT scores and right-mPFC/ rostral ACC DLPFC
Seok et al., (2014)	IA: 15 (15;-) (M=22,20; SD=3,07) CG: 15 (15;-) (M=22,47; SD= 2,53)	IAS K Scale	Whole brain	Decreased FC between the OFC-inferior parietal cortex, OFC-putamen, OFC-ACC, Insula - ACC, and amygdala-insula in IA. Stronger negative correlation between FC OFC-insula in IA. No significant relationship between FC strength and the degree of IA and degree of impulsivity was seen.

Note: PIU-Q: Problematic Internet Use Questionnaire; BSMAS: Bergen Social Media Addiction Scale; SABAS: Smartphone Application-Based Addiction Scale; IAT: Internet Addiction Test; SAPS: Smartphone Addiction Proneness Scale; CIAS-R: Chen internet addiction scale-revised questionnaire; IAS: Internet Addiction Scale; K-SCale: Korean-version Internet Addiction Self-Diagnosis Scale.

Table 3
Task-based fMRI in Adolescents.

ARTICLE	SAMPLE (M; F)	IA SCALE	TASK	RESULTS
Kim et al. (2014)	IA: 15 (M=13,87; SD=0,83) GC: 15 (M=13,87; SD=0,83)	K-AIAS	Discrimination task: answer if the symbol was on the right or left. Could receive feedback, money, or no reward.	Greater activation of the DLPFC, and negative correlation between activation of the left superior temporal and time spent on the internet.
Li et al. (2014)	IA: 18 (M= 15,1; SD=1,4) GC: 23 (M=15,2; SD=0,5)	YDQ	Go-Nogo	Ineffective connectivity in frontal-basal ganglia pathway by response inhibition in IA participants

Note: K-AIAS: Korean Adolescent Internet Addiction Scale; YDQ: Young's Diagnostic Questionnaire.

left lateral PFC, the right mPFC and lateral parietal (Siste et al., 2022) was found in adolescents with IA.

3.3.2. *Young adults*

Seven studies have reported abnormal activity in different brain regions using resting-state fMRI associated with rewarding system and executive function in young adults with IA, two of which also included specific SPA testing.

Differences between young and older adults at high risk of developing IA disorders had been reported. IA was associated with abnormal resting state activity in some brain regions such as dorsolateral prefrontal cortex, cingulate gyrus, insula, amygdala, and cerebellum. Occipital-putamen FC was increased (Patil et al., 2021) and decreased between OFC-putamen in IA (Seok, Sohn, 2014). Also, a decrease in functional connectivity and integrity between the amygdala and

Table 4
Task-based fMRI in adults.

ARTICLE	SAMPLE (M; F)	IA Scale	TASK	RESULTS
Yoon et al. (2015)	IA: 11 (10;1) (M=25,45; SD=3,23) GC: 22 (11;11) (M=23,91; SD=2,52)	IAT	Monetary incentive learning task	Increased activity in the posterior insula during the gratification condition and decreased during avoidance.
Seok et al. (2015)	IA: 15 (15;-) (M=22,20; SD=3,07) CG: 15 (15;-) (M=22,47; SD=2,53)	IAS K Scale	Financial decision-making Task	More frequent risky decision making; greater activation in the dorsal ACC and the left caudate nucleus and less activation in the ventrolateral PFC in IA.
Darnai et al. (2019)	IA: 60 (30;30) (M=22,0; SD=2,08)	PIUQ	Stroop	Significant deactivations in areas related to the DMN (precuneus, posterior cingulate gyrus), negatively correlated with PIUQ during incongruent stimuli. Positive correlation with PIUQ in inhibitory control network (left inferior frontal gyrus, left frontal pole, left central opercular, left frontal opercular, left OFC and left insular cortex).
Dong et al. (2014)	IA: 15 (M=21,2; SD= 3,2) GC: 15 (M= 22,1; SD=3,6)	IAT	Stroop	Greater activation of the superior temporal gyrus in change of activity. In the difficult-easy greater activation of the bilateral insula; in the easy-difficult bilateral precuneus.
Dong et al. (2013)	IA: 15 (M=23,8; SD= 3,7) GC: 15 (M= 24,1; SD= 3,3)	IAT	Stroop	Greater activation of ACC and lower OFC activation in error-monitoring
Dieter et al. (2017)	IA: 25 (17;8) (M=25,60; SD=5,93) GC: 23 (10;13) (M=26,13; SD=5,49)	AICA 30 OSVe-S	Go-Nogo Stroop	No differences were found in Go/no go or Stroop in ACC activation. Exploratory analysis: decrease activation of the superior and middle left temporal in IA, more in video games than in smartphone.
Darnai et al. (2022)	IA: 60 (30; 30) (M=22,0; SD=2,08)	PIUQ	Silent word generation task	DMN was altered in IA during the task. FC Broca's area showed altered with other language network and occipital areas in IA.
Arató et al. (2023)	IA: 65 (32; 33) (M=22,50; SD=2,68)	PIUQ SABAS	Facial Emotion Recognition Paradigm	FC between the right amygdala and frontal regions (related to emotional cognitive control and social brain networks) positively correlated with IA and SPA.

Note: PIU-Q: Problematic Internet Use Questionnaire; SABAS: Smartphone Application-Based Addiction Scale; IAT: Internet Addiction Test; AICA 30: Assessment of Internet and Computer game Addiction; OSVe-S: online addictive behavior; IAS: Internet Addiction Scale; K-SCale: Korean-version Internet Addiction Self-Diagnosis Scale.

anterior cingulate cortex (ACC) (Cheng and Liu, 2020) was found. Similarly Li et al. (2015) found a significant association between ACC atrophy and Internet Addiction Test (IAT) scores (high scores on the test were related to low connectivity of ACC). Young adults with IA show decreased FC between Insula-ACC, and amygdala-insula and negative correlation between OFC-insula connectivity (Seok, Sohn, 2014). Abnormal amygdala-thalamus functional connectivity was also found, suggesting emotional dysregulation associated with IA. Young adults with IA were found to have a reduced amygdala-ACC FC and an increased FC between the amygdala, precuneus, and superior occipital gyrus (SOG) (Cheng and Liu, 2020). Furthermore, Li et al. (2015) found a low negative correlation between IAT and right medial prefrontal cortex (mPFC)/rostral ACC and DLPFC scores. These results suggest that people with high IAT scores had lower activity in the right mPFC.

Another frontal region that has been implicated in IA is the DLPFC. It has been found that despite the greater FC in the right DLPFC (Wang et al., 2019), there is an association between atrophy in this region and IAT scores (the higher the test score, the greater the DLPFC atrophy) (Li et al., 2015), and reduced amygdala-DLPFC functional connectivity in IA compared to controls (Cheng and Liu, 2020), suggesting that top-down control of attention involving executive control is reduced (Lee et al., 2021). In addition, impairments have also been found in some other brain regions, such as the parietal lobe, where participants with IA showed reduced connectivity between the OFC-inferior parietal cortex compared to controls (Seok, Sohn, 2014), increased FC in the amygdala-precuneus-superior occipital gyrus (SOG) in IA (Cheng et al., 2020), and increased connectivity in the intraparietal sulcus-right middle occipital gyrus in the dorsal pathway in SPA (Lee et al., 2021). Young adults with IA had shown an increase in FC in the occipital-putamen (associated with lower effectiveness in cognitive control and decision making) (Patil et al., 2021) and a decrease in FC in the calcarine and lingual gyrus (Wang et al., 2019). Reduced FC between the occipital pole network and the auditory network was also found in young adults with SPA (Áfra et al., 2023). Although Patil et al. (2021) reported a disruption of executive cerebellar networks in young adults, Wang et al. (2019) did not find this reduction, and it did not occur in older adults with IA (Patil et al., 2021). In addition, an association was found between IAT score and altered FC in the left parahippocampal gyrus and bilateral superior temporal pole (Wang et al., 2019).

3.4. TASK-BASED fMRI

3.4.1. Adolescents

Using task-based fMRI, two studies have shown functional alterations in adolescents with IA. One study used the Go - Nogo task, which is a cognitive task that requires participants to respond by pressing a button when they see a Go condition (i.e. a green dot) and a Nogo condition in which no response is required (i.e. a red dot appears on the screen). Ineffective connectivity in the frontal-basal ganglia pathway due to response inhibition in the Go-Nogo task condition was found in participants with IA (Li et al., 2014). The other study used a monetary discrimination task in fMRI in which participants had to answer whether an abstract symbol was on the right or left side of the screen and received money if they answered correctly. Participants with IA who performed a monetary discrimination task in fMRI showed increased activation in the DLPFC compared to control participants (Kim et al., 2014). Adolescents with IA had reduced reward sensitivity regardless of the type of reward and feedback given (Kim et al., 2014).

3.4.2. Young adults

Eight studies have demonstrated functional abnormalities in young adults with IA using task-based fMRI, one of these eight also included specific SPA tests. Participants with IA showed increased activation in posterior insula in gain trials and decreased in loss trials implicating enhanced reward sensitivity and decreased loss sensitivity (Yoon et al., 2015) and more frequent risky decision making (Seok et al., 2015). Young adults with IA showed differences in neural activity in ACC, left caudate nucleus, inferior, ventrolateral, and medial frontal gyrus as well as cerebellar and occipital regions during a cue vs. neutral stimulus condition. Moreover, three studies with Stroop tasks while undergoing a fMRI was identified. Stroop's task evaluates selective attention and inhibitory control. People with high scores in IA had greater activation of the ACC and less activation of the OFC than controls before the Stroop task (Dong et al., 2013, 2014) and greater activation in the superior temporal gyrus when participants changed condition on the task. Specifically, the bilateral insula was involved in the change of condition from difficult to easy and the precuneus when the change was made from easy to difficult task (Dong et al., 2014). Darnai et al. (2019) found negative correlations between PIUQ scores and DMN regions during incongruent stimuli, and positive correlations with inhibitory control networks in the left inferior frontal gyrus, left frontal pole, left central opercular, left frontal opercular, left frontal orbital, and left insular cortex.

No differences in ACC activation were found in a Go-Nogo task in young adults with IA. However, when comparing SPA and gaming, a decrease in superior temporal activation and left medial temporal activation was found in people with high scores in gaming addiction, but not in SPA (Dieter et al., 2017). Some functional impairments were found in Broca's area with other language networks and occipital regions during the silent word generation task in young adults with IA (Darnai et al., 2022). In addition, Arató et al. (2023) found a positive association between IA and SPA test scores with increased FC in amygdala and frontal regions related to emotional cognitive control and social brain networks.

4. Discussion

The aim of this systematic review was to provide an overall synthesis of the literature on differential adverse effects on cognition in young adults and adolescents with IA and SPA using fMRI. Although IA and SPA may appear similar, as they refer to the full range of activities that can be performed via the Internet, some researchers have suggested the importance of distinguishing between IA and SPA, as they may have a different effects on the development or maintenance of behavioral addictions and different patterns of addiction risk factors (Jeong et al., 2019; Montag et al., 2021). To our knowledge, this is the first review comparing the effects of IA and SPA on executive functions and the reward system in adolescents and young adults using fMRI. The studies presented in this review had highlighted that people at high risk of IA in particular had impaired executive functions and had experimented changes in the reward system. Specifically, the reward system has been affected by excessive Internet and smartphone use in brain regions including the striatum, amygdala, insula, and ACC.

Studies in this review have found that young adults and adolescents with IA exhibit reduced ACC activity. The ACC is closely related to the interaction between cognitive control and reward-related networks, which may be a critical area involved in the addiction process (Gläscher et al., 2012). Furthermore, in adolescents with high scores on IA, Wang et al. (2017) found reduced connectivity in the right fronto-parietal circuit, mPFC and anterior DMN. In young adults with IA a reduction in connectivity between IA and ACC was observed (Cheng and Liu, 2020), similar to the findings of Li et al. (2015), who found a significant association between ACC atrophy and IAT scores (high scores on the test were associated with low connectivity in the ACC). In a Stroop task, Dong et al. (2013) observed greater activation of the ACC and less activation of the orbitofrontal cortex in young adults with high scores on

IA tests.

Reduced amygdala connectivity has been shown in adults with IA and SPA. Young adults with IA were found to have reduced FC between the amygdala and ACC, as well as increased FC between the amygdala and precuneus-SOG and between the amygdala and insula (Cheng and Liu, 2020; Seok, Sohn, 2014). Abnormal FC in the amygdala has implications for self-control, which is critical for the regulation of craving and plays an important role in the development and maintenance of addiction (Crunelle et al., 2015). Abnormal functional connectivity between the amygdala and thalamus has also been found, suggesting emotional dysregulation associated with IA (Cheng and Liu, 2020). Arató et al. (2023) found increased FC in the amygdala and frontal regions during the Facial Emotion Recognition Paradigm, which was positively associated with both IA and SPA test scores. The amygdala is the main center of emotional control. Its connections not only produce an emotional response, but it is also involved in cognitive inhibition.

The insula's involvement in the integration of emotion and cognition contributes to addiction and the phenomenon of craving (Garavan, 2010). Abnormalities in this region may lead to impaired self-awareness and emotion regulation in IA. Some impairments have been found in FC between the insula and other brain regions. Specifically, young adults with IA show decreased FC between insula-ACC and amygdala-insula and a negative correlation between OFC-insula (Seok, Sohn, 2014). On the other hand, in adolescents with IA, Yoon et al. (2015) showed increased activity in gain trials and decreased activity in loss trials, suggesting increased reward sensitivity and decreased loss sensitivity, which is consistent with the data of decreased reward sensitivity reported by Kim et al. (2014). This behavioral performance suggests impaired risk assessment, which may be the reason why people with IA continue to overuse technological devices despite the risks of widely known negative consequences, a mechanism also typical of individuals with substance use and pathological gambling disorders (Ford, Håkansson, 2020). Despite these findings, studies in adolescents (Siste et al., 2022) and young adults (Dong et al., 2014; Yoon et al., 2015) have reported increased connectivity in the posterior and anterior insula in IA. It is not clear why these differences exist, but it is possible that differences such as the task they were exposed to or the small sample size may play a role.

Adolescents and young adults have also shown heterogeneous neural activity associated with executive function. Although greater FC was found in the right DLPFC (Wang et al., 2019), recent studies have found a reduction in DLPFC functional connectivity in young adults with IA, suggesting that top-down control of attention, including executive control, is reduced (Cheng and Liu, 2020; Lee et al., 2021). Adolescents with high IA scores have been found to have reduced FC in cortico-subcortical circuits (prefrontal and parietal) (Hong et al., 2013), in the right fronto-parietal circuit, mPFC and anterior DMN connectivity compared to controls. More specifically, Li et al. (2015) found an association between DLPFC atrophy and IAT scores (the higher the test score, the greater the DLPFC atrophy). In the same line, Siste et al. (2022) found a reduction in FC in the left lateral PFC - right mFC and greater functional connectivity in the left lateral PFC - left anterior insula in adolescents with IA. These findings have been explained in adolescents as ineffective connectivity between frontal-basal ganglia pathways due to response inhibition (Li et al., 2014). This inability to control their internet use is strongly related to inhibitory control (Wang et al., 2020) and emotional regulation deficit (Hormes et al., 2014; Hou et al., 2019), both processes related to central executive function. Dysfunction in these (pre-)frontal regions has been shown to contribute to the development of symptoms characteristic of substance addiction, such as craving, denial of addiction, and increased compulsive use (Goldstein and Volkow, 2011).

In addition to the frontal lobe, changes were found in the parietal and occipital lobes. Compared to controls, young adults with IA had decreased FC between the OFC-inferior parietal cortex (Seok, Sohn, 2014), stronger connectivity in the intraparietal sulcus - right middle occipital gyrus in the dorsal pathway (Lee et al., 2021) and increased functional connectivity in the amygdala-precuneus-superior occipital gyrus (SOG) (Cheng et al., 2020). Some impairments in Broca's area have been found with other language networks and occipital regions during the Silent Word Generation Task (Darnai et al., 2022). Furthermore, these impairments were also found in adolescents with IA, specifically a reduction in FC of the lateral parietal PFC (Siste et al., 2022). Young adults with IA had shown an increase in FC in the occipital-putamen (associated with reduced effectiveness in cognitive control and decision-making) (Patil et al., 2021) and a reduction in FC in the calcarine, lingual gyrus (Wang et al., 2019) and between the occipital pole network and the auditory network (Áfra et al., 2023). Hyperactivation in the occipital lobe has also been associated with substance dependence (Qiu and Wang, 2021).

In general, the studies in this review have shown differences in functional activity in people with excessive internet and smartphone use compared to controls. However, there are also discrepancies between the results found, and these findings should be taken with caution due to several limitations: (i) The included studies in this review were heterogeneous in terms of scanners and image processing procedures, such as smoothing, registration or imaging modalities, and data analysis, such as significance, techniques for correcting multiple comparisons and statistical power, which makes it difficult to make balanced comparisons between studies. (ii) The correlational results presented in the studies reviewed here do not provide a complete explanation of how IA/SPA might alter the regions of the brain. Longitudinal studies are needed to clarify the causal relationship between brain functional abnormalities and the incidence of IA and SPA. (iii) Of the 21 articles included in this review, only three included tests to assess SPA, and these were all in young adults. Due to the limited number of studies, it was not possible to conduct a detailed analysis of the differences in behavioral addiction. (iv) The studies included in this review assessed IA and SPA on the basis of excessive use or consequences of daily life activities using self-report measures. This type of measurement has inherent limitations, the most important being possible response bias and social desirability. (v) It should be noted that the review included papers that assessed Internet use as an addictive or problematic behavior, so it would be interesting in the future to reach a consensus on the conceptual boundaries between addiction and problematic behavior, as well as between IA and SPA. (vi) To clarify the effect on cognitive abilities, more studies with different task conditions in IA and SPA are needed.

The search for the biological basis of mental disorders, rather than deepening the reductionist model, has led to the proposal of an alternative framework for the analysis of mental disorders. Despite the criticisms received by the biopsychosocial model (Bolton, Gillett, 2019), it is proposed that mental disorders form a network that includes biological and psychological factors, without assuming an explanatory priority. Mental disorders are described as networks of symptoms linked by a causal relationship that precludes the identification of a common cause of symptomatology with a neurobiological condition, and it is recognised that the relationships of symptom networks depend on social, cultural and historical contexts (Borsboom, Cramer and Kalis, 2018). IA and SPA can be understood as disorders that combine brain and contextual aspects, as proposed by the biopsychosocial model. The understanding and treatment of behavioral addictions through the biopsychosocial lens allows for a comprehensive and individualized approach. It recognises that these addictions are not the result of one factor alone, but rather a complex interplay of biological, psychological and social elements.

There is a need for an international protocol for the evaluation of instruments for the assessment of the effects of behavioral dependence on neuroanatomical changes. A standardized procedure for assessing the effects of these types of addiction on different abilities could improve the accuracy of these measures while increasing the generalisability of the findings. Although we aimed to identify the differences and similarities between IA and SPA, this was not possible because the studies identified

used both terms interchangeably and did not differentiate between IA and SPA outcomes. The findings suggest that adolescents and young adults with higher levels of IA and SPA tend to have impaired executive function, including a set of cognitive processes (adaptive thinking, planning, self-monitoring, self-control, working memory, time management, and organization) (Lin et al., 2018; Zhang et al., 2023). Further research is needed to clarify the effects of IA and SPA on cognitive and psychological states and their possible brain impairments, as well as new forms of assessment and treatment for these behavioral addictions. In addition, it would be interesting for future research to investigate the cognitive effects produced by other forms of technology addiction in order to provide a more in-depth analysis of these types of behavioral addictions and to draw clear conclusions about the neural correlates and cognitive effects associated with each addiction.

5. Conclusions

IA and SPA have been studied extensively, with a substantial body of literature examining their effects on mental health and cognition. However, to our knowledge, this is the first review to compare the effects of IA and SPA on executive function and the reward system in adolescents and young adults using fMRI. In general, neuroimaging studies have revealed an emerging pattern of generalized changes in brain activity in people with IA and SPA. Overall, young adults and adolescents with IA and SPA show functional and resting-state brain abnormalities compared to controls, including functional impairments in reward (striatum, AAC, insula, amygdala) and executive functions (DLPFC, frontal/parietal/occipital lobes). The current evidence may help to identify functional impairments associated with IA and SPA and highlights the neuroanatomical similarities and differences between these behavioral addictions.

Declaration

This study was a systematic review of the previously published studies and did not use original human or animal data.

Funding

The authors declare that this research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of Competing Interest

No conflicts of interest, financial or otherwise, are declared by the authors.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.neubiorev.2024.105572.

References

Áfra, E., Janszky, J., Perlaki, G., Orsi, G., Nagy, S.A., Arató, Á., Szente, A., Alhour, H.A. M., Kis-Jakab, G., Darnai, G., 2023. Altered functional brain networks in problematic Internet use: resting-state fMRI study. Res. Sq. (Res. Sq.). https://doi.org/10.21203/rs.3.rs-2718429/v1.

Alavi, S.S., Ferdosi, M., Jannatifard, F., Eslami, M., Alaghemandan, H., Setare, M., 2012. Behavioral addiction versus substance addiction: correspondence of psychiatric and psychological views. Int. J. Prev. Med. 3 (4), 290–294.

Altbäcker, A., Plózer, E., Darnai, G., Perlaki, G., Horváth, R., Orsi, G., Nagy, S.A., Bogner, P., Schwarcz, A., Kovács, N., Komoly, S., Clemens, Z., Janszky, J., 2016. Problematic internet use is associated with structural alterations in the brain reward system in females. Brain Imaging Behav. 10 (4), 953–959. https://doi.org/10.1007/s11682-015-9454-9.

Arató, Á., Nagy, S.A., Perlaki, G., Orsi, G., Szente, A.T., Kis-Jakab, G., Áfra, E., Alhour, H. A., Kovács, N., Janszky, J., Darnai, G., 2023. Emotional face expression recognition in problematic Internet use and excessive smartphone use: task-based fMRI study. Sci. Rep. 13 (1), 354. https://doi.org/10.1038/s41598-022-27172-0.

Ballestar-Tarín, M.L., Simó-Sanz, C., Chover-Sierra, E., Saus-Ortega, C., Casal-Angulo, M. D.C., Martínez-Sabater, A., 2020. Self-perception of dependence as an indicator of smartphone addiction-establishment of a cutoff point in the SPAI-Spain inventory. Int. J. Environ. Res. Public Health 17 (11), 3838. https://doi.org/10.3390/ijerph17113838.

Bari, A., Robbins, T.W., 2013. Inhibition and impulsivity: behavioral and neural basis of response control. Prog. Neurobiol. 108, 44–79. https://doi.org/10.1016/j.pneurobio.2013.06.005.

Bates, M.E., Pawlak, A.P., Tonigan, J.S., Buckman, J.F., 2006. Cognitive impairment influences drinking outcome by altering therapeutic mechanisms of change. Psychol. Addict. Behav.: *J. Soc. Psychol. Addict. Behav.* 20 (3), 241–253. https://doi.org/10.1037/0893-164X.20.3.241.

Billieux, J., Maurage, P., Lopez-Fernandez, O., Kuss, D.J., Griffiths, M.D., 2015. Can disordered mobile phone use be considered a behavioral addiction? An update on current evidence and a comprehensive model for future research. Curr. Addict. Rep. 2 (2), 156–162. https://doi.org/10.1007/s40429-015-0054-y.

Bolton, D., & Gillett, G. (2019). *The Biopsychosocial Model of Health and Disease: New Philosophical and Scientific Developments.* Palgrave Pivot.

Borsboom, D., Cramer, A.O.J., Kalis, A., 2018. Brain disorders? Not really: why network structures block reductionism in psychopathology research. Behav. Brain Sci. 42, e2 https://doi.org/10.1017/S0140525X17002266.

Brand, M., Young, K.S., Laier, C., Wölfling, K., Potenza, M.N., 2016. Integrating psychological and neurobiological considerations regarding the development and maintenance of specific Internet-use disorders: an interaction of Person-Affect-Cognition-Execution (I-PACE) model. Neurosci. Biobehav. Rev. 71, 252–266. https://doi.org/10.1016/j.neubiorev.2016.08.033.

Brand, M., Wegmann, E., Stark, R., Müller, A., Wölfling, K., Robbins, T.W., Potenza, M. N., 2019. The Interaction of Person-Affect-Cognition-Execution (I-PACE) model for addictive behaviors: update, generalization to addictive behaviors beyond internet-use disorders, and specification of the process character of addictive behaviors. Neurosci. Biobehav. Rev. 104, 1–10. https://doi.org/10.1016/j.neubiorev.2019.06.032.

Cabañas, M., & Korzeniowski, C. (2015). Uso de celular e Internet: su relación con planificación y control de la interferencia. *Revista Argentina de Ciencias del Comportamiento*, 7(1), 5–16. https://www.redalyc.org/articulo.oa?id=333439929002.

Cai, Z., Mao, P., Wang, Z., Wang, D., He, J., Fan, X., 2023. Associations between problematic internet use and mental health outcomes of students: a meta-analytic review. Adolesc. Res. Rev. 8 (1), 45–62. https://doi.org/10.1007/s40894-022-00201-9.

Cao, F., Su, L., Liu, T., Gao, X., 2007. The relationship between impulsivity and Internet addiction in a sample of Chinese adolescents. Eur. Psychiatry.: J. Assoc. Eur. Psychiatr. 22 (7), 466–471. https://doi.org/10.1016/j.eurpsy.2007.05.004.

Cheng, C., Lau, Y.C., Chan, L., Luk, J.W., 2021. Prevalence of social media addiction across 32 nations: meta-analysis with subgroup analysis of classification schemes and cultural values. Addict. Behav. 117, 106845 https://doi.org/10.1016/j.addbeh.2021.106845.

Cheng, H., Liu, J., 2020. Alterations in amygdala connectivity in internet addiction disorder. Sci. Rep. 10 (1) https://doi.org/10.1038/s41598-020-59195-w.

Choi, S.W., Kim, D.J., Choi, J.S., Ahn, H., Choi, E.J., Song, W.Y., Kim, S., Youn, H., 2015. Comparison of risk and protective factors associated with smartphone addiction and Internet addiction. J. Behav. Addict. 4 (4), 308–314. https://doi.org/10.1556/2006.4.2015.043.

Crone, E.A., Steinbeis, N., 2017. Neural perspectives on cognitive control development during childhood and adolescence. Trends Cogn. Sci. 21 (3), 205–215. https://doi.org/10.1016/j.tics.2017.01.003.

Curtis, C.E., D'Esposito, M., 2003. Persistent activity in the prefrontal cortex during working memory. Trends Cogn. Sci. 7 (9), 415–423. https://doi.org/10.1016/s1364-6613(03)00197-9.

Daniyal, M., Javaid, S.F., Hassan, A., Khan, M.A.B., 2022. The relationship between cellphone usage on the physical and mental wellbeing of university students: a cross-sectional study. Int. J. Environ. Res. Public Health 19 (15), 9352. https://doi.org/10.3390/ijerph19159352.

Darnai, G., Perlaki, G., Zsido, A.N., Inhof, O., Orsi, G., Horváth, R., Nagy, S., Lábadi, B., Tényi, D., Kovács, N., Dóczi, T., Demetrovics, Z., Janszky, J., 2019. Internet addiction and functional brain networks: task-related fMRI study. Sci. Rep. 9 (1) https://doi.org/10.1038/s41598-019-52296-1.

Darnai, G., Perlaki, G., Orsi, G., Arató, Á., Szente, A., Horváth, R., et al., 2022. Language processing in Internet use disorder: task-based fMRI study. PLoS ONE 17 (6), e0269979. https://doi.org/10.1371/journal.pone.0269979.

Demirci, K., Orhan, H., Demirdaş, A., Akpınar, A., Sert, H., 2014. Validity and reliability of the Turkish version of the smartphone addiction scale in a younger population. Klin. Psikofarmakol. *Bul. -Bull. Clin. Psychopharmacol.,* 24 (3), 226–234. https://doi.org/10.5455/bcp.20140710040824.

Dieter, J., Hoffmann, S., Mier, D., Reinhard, I., Beutel, M., Vollstädt-Klein, S., Kiefer, F., Mann, K., Leménager, T., 2017. The role of emotional inhibitory control in specific internet addiction – an fMRI study. Behav. Brain Res. 324, 1–14. https://doi.org/10.1016/j.bbr.2017.01.046.

Dong, G., Shen, Y., Huang, J., Du, X., 2013. Impaired error-monitoring function in people with internet addiction disorder: an event-related fMRI study. Eur. Addict. Res. 19 (5), 269–275. https://doi.org/10.1159/000346783.

Dong, G., Lin, X., Zhou, H., Lu, Q., 2014. Cognitive flexibility in internet addicts: fMRI evidence from difficult-to-easy and easy-to-difficult switching situations. Addict. Behav. 39 (3), 677–683. https://doi.org/10.1016/j.addbeh.2013.11.028.

Ford, M., Håkansson, A., 2020. Problem gambling, associations with comorbid health conditions, substance use, and behavioural addictions: opportunities for pathways to treatment. PloS One 15 (1), e0227644. https://doi.org/10.1371/journal.pone.0227644.

Gao, Q., Jia, G., Zhao, J., Zhang, D., 2019. Inhibitory control in excessive social networking users: evidence from an event-related potential-based go-nogo task. Front. Psychol. 10, 1810. https://doi.org/10.3389/fpsyg.2019.01810.

Garavan, H., 2010. Insula and drug cravings. Brain Struct. Funct. 214 (5-6), 593–601. https://doi.org/10.1007/s00429-010-0259-8.

Gläscher, J., Adolphs, R., Damasio, H., Bechara, A., Rudrauf, D., Calamia, M., Paul, L.K., Tranel, D., 2012. Lesion mapping of cognitive control and value-based decision making in the prefrontal cortex. Proc. Natl. Acad. Sci. 109 (36), 14681–14686. https://doi.org/10.1073/pnas.1206608109.

Goldstein, R.Z., Volkow, N.D., 2011. Dysfunction of the prefrontal cortex in addiction: neuroimaging findings and clinical implications. Nat. Rev. *Neurosci.* 12 (11), 652–669. https://doi.org/10.1038/nrn3119.

Hong, S.H., Zalesky, A., Cocchi, L., Fornito, A., Choi, E.H., Kim, H., Suh, J., Kim, C.D., Kim, J.N., Yi, S., 2013. Decreased functional brain connectivity in adolescents with internet addiction. PLOS ONE 8 (2), e57831. https://doi.org/10.1371/journal.pone.0057831.

Horvath, J., Mundinger, C., Schmitgen, M.M., Wolf, N.D., Sambataro, F., Hirjak, D., Kubera, K.M., Koenig, J., Wolf, R., 2020. Structural and functional correlates of smartphone addiction. Addict. Behav. 105, 106334 https://doi.org/10.1016/j.addbeh.2020.106334.

Ioannidis, K., Hook, R., Goudriaan, A.E., Vlies, S., Fineberg, N.A., Grant, J.E., Chamberlain, S.R., 2019. Cognitive deficits in problematic internet use: meta-analysis of 40 studies. Br. J. Psychiatry.: J. Ment. Sci. 215 (5), 639–646. https://doi.org/10.1192/bjp.2019.3.

Jarmolowicz, D.P., Mueller, E.T., Koffarnus, M.N., Carter, A.C., Gatchalian, K.M., & Bickel, W.K. (2013). Executive Dysfunction in Addiction. *Wiley-Blackwell eBooks* (pp. 27–61). https://doi.org/10.1002/9781118384404.ch2.

Jeong, Y.J., Suh, B., Gweon, G., 2019. Is smartphone addiction different from internet addiction? Comparison of addiction-risk factors among adolescents. Behav. Inf. Technol. 39 (5), 578–593. https://doi.org/10.1080/0144929x.2019.1604805.

Kim, J., Son, J., Choi, W.S., Kim, Y., Oh, J.J., Lee, S., Kim, J.K., 2014. Neural responses to various rewards and feedback in the brains of adolescent Internet addicts detected by functional magnetic resonance imaging. Psychiatry Clin. Neurosci. 68 (6), 463–470. https://doi.org/10.1111/pcn.12154.

Kim, S.J., Marsch, L.A., Hancock, J.T., Das, A.K., 2017. Scaling Up Research on Drug Abuse and Addiction Through Social Media Big Data. J. Med. Internet Res. 19 (10), e353 https://doi.org/10.2196/jmir.6426.

Kuss, D.J., Griffiths, M.D., 2017. Social networking sites and addiction: ten lessons learned. Int. J. Environ. Res. Public Health 14 (3), 311. https://doi.org/10.3390/ijerph14030311.

Kuss, D.J., Griffiths, M.D., Karila, L., Billieux, J., 2014. Internet addiction: a systematic review of epidemiological research for the last decade. Curr. Pharm. Des. 20 (25), 4026–4052. https://doi.org/10.2174/13816128113199990617.

Kwak, Y., Kim, H., Ahn, J., 2022. Impact of Internet usage time on mental health in adolescents: using the 14th Korea Youth Risk Behavior Web-Based Survey 2018. PLoS One 17 (3), e0264948. https://doi.org/10.1371/journal.pone.0264948.

Kwon, M., Kim, D.J., Cho, H., Yang, S., 2013. The smartphone addiction scale: development and validation of a short version for adolescents. PloS One 8 (12), e83558. https://doi.org/10.1371/journal.pone.0083558.

Lee, D., Lee, J., Namkoong, K., Jung, Y., 2021. Altered functional connectivity of the dorsal attention network among problematic social network users. Addict. Behav. 116, 106823 https://doi.org/10.1016/j.addbeh.2021.106823.

Li, B., Friston, K.J., Liu, J., Liu, Y., Zhang, G., Cao, F., Su, L., Yao, S., Lu, H., Hu, D., 2014. Impaired frontal-basal ganglia connectivity in adolescents with internet addiction. Sci. Rep. 4, 5027. https://doi.org/10.1038/srep05027.

Li, W., Li, Y., Yang, W., Zhang, Q., Wei, D., Li, W., Hitchman, G., Qiu, J., 2015. Brain structures and functional connectivity associated with individual differences in Internet tendency in healthy young adults. Neuropsychologia 70, 134–144. https://doi.org/10.1016/j.neuropsychologia.2015.02.019.

Logan, G., 1988. Toward an instance theory of automatization. Psychol. Rev. 95, 492–527. https://doi.org/10.1037/0033-295X.95.4.492.

Luria, A.R., 1973. The Working Brain. Basic Books, New York.

Marciano, L., Camerini, A., Schulz, P.J., 2020. Neuroticism in the digital age: a meta-analysis. Comput. Hum. Behav. Rep. 2, 100026 https://doi.org/10.1016/j.chbr.2020.100026.

Meng, S.Q., Cheng, J.L., Li, Y.Y., Yang, X.Q., Zheng, J.W., Chang, X.W., Shi, Y., Chen, Y., Lu, L., Sun, Y., Bao, Y.P., Shi, J., 2022. Global prevalence of digital addiction in general population: A systematic review and meta-analysis. Clin. Psychol. Rev. 92, 102128 https://doi.org/10.1016/j.cpr.2022.102128.

Montag, C., Wegmann, E., Sariyska, R., Demetrovics, Z., Brand, M., 2021. How to overcome taxonomical problems in the study of Internet use disorders and what to do with "smartphone addiction"? J. Behav. Addict. 9 (4), 908–914. https://doi.org/10.1556/2006.8.2019.59.

Nikhita, C.S., Jadhav, P.R., Ajinkya, S.A., 2015. Prevalence of mobile phone dependence in secondary school adolescents. VC06–VC09 *Journal Clin. Diagn. Res.*: JCDR 9 (11). https://doi.org/10.7860/JCDR/2015/14396.6803.

Olson, J.A., Sandra, D.A., Colucci, É.S., Bikaii, A.A., Chmoulevitch, D., Nahas, J., Raz, A., Veissière, S.P.L., 2022. Smartphone addiction is increasing across the world: a meta-analysis of 24 countries. Comput. Hum. Behav. 129, 107138 https://doi.org/10.1016/j.chb.2021.107138.

Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., McGuinness, L.A., Moher, D., 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ (Clin. Res. Ed.) 372, n71. https://doi.org/10.1136/bmj.n71.

Patil, A.P., Madathil, D., Huang, C., 2021. Age-related and individual variations in altered prefrontal and cerebellar connectivity associated with the tendency of developing internet addiction. Hum. Brain Mapp. 42 (14), 4525–4537. https://doi.org/10.1002/hbm.25562.

Pies R. (2009). Should DSM-V Designate "Internet Addiction" a Mental Disorder?. Psychiatry (Edgmont (Pa.: Township)), 6(2), 31–37.

Ponnusamy, S., Iranmanesh, M., Foroughi, B., Hyun, S.S., 2020. Drivers and outcomes of Instagram addiction: psychological well-being as moderator. Comput. Hum. Behav. 107, 106294 https://doi.org/10.1016/j.chb.2020.106294.

Pontes, H., Griffiths, M.D., 2014. Assessment of internet gaming disorder in clinical research: past and present perspectives. Clin. Res. Regul. Aff. 31 (2-4), 35–48. https://doi.org/10.3109/10601333.2014.962748.

Qiu, Z., Wang, J., 2021. A voxel-wise meta-analysis of task-based functional MRI studies on impaired gain and loss processing in adults with addiction. *J. Psychiatry Neurosci.*: JPN 46 (1), E128–E146. https://doi.org/10.1503/jpn.200047.

Rosselli, M., Christopher, D.M., 2019. Executive dysfunctions associated with the use of information technology. Springer eBooks,, pp. 177–197. https://doi.org/10.1007/978-3-030-25077-5_9.

Salama, B., 2020. Prevalence and associated factors of Internet addiction among undergraduate students at Al-Beheira Governorate, Egypt. Int. J. Public Health 65 (6), 905–910. https://doi.org/10.1007/s00038-020-01429-9.

Salehi, M., Abbaspour, Z., Molana, A., Shahini, N., 2023. Impulsivity, inhibition, and internet addiction in medical students of North of Iran. Front. Psychiatry 13, 1002625. https://doi.org/10.3389/fpsyt.2022.1002625.

Schoenbaum, G., Roesch, M.R., Stalnaker, T.A., 2006. Orbitofrontal cortex, decision-making and drug addiction. Trends Neurosci. 29 (2), 116–124. https://doi.org/10.1016/j.tins.2005.12.006.

Seki, T., Hamazaki, K., Natori, T., Inadera, H., 2019. Relationship between internet addiction and depression among Japanese university students. J. Affect. Disord. 256, 668–672. https://doi.org/10.1016/j.jad.2019.06.055.

Seok, J., Sohn, J., 2014. Altered functional disconnectivity in internet addicts with resting-state functional magnetic resonance imaging. J. Ergon. Soc. Korea 33 (5), 377–386. https://doi.org/10.5143/jesk.2014.33.5.377.

Seok, J.W., Lee, K.H., Sohn, S., Sohn, J.H., 2015. Neural substrates of risky decision making in individuals with Internet addiction. Aust. N. Z. J. Psychiatry 49 (10), 923–932. https://doi.org/10.1177/0004867415598009.

Shafiee-Kandjani, A.R., Mohammadzadeh, Z., Amiri, S., Asghar, A., Sarbakhsh, P., Salman, S., 2019. Attention, working memory and executive functions in patients with internet addiction disorder. J. Inj. Violence Res. 11 (4 Suppl 2). Paper No. 70.

Shaw, M., Black, D.W., 2008. Internet addiction: definition, assessment, epidemiology and clinical management. CNS Drugs 22 (5), 353–365. https://doi.org/10.2165/00023210-200822050-00001.

Siste, K., Pandelaki, J., Miyata, J., Oishi, N., Tsurumi, K., Fujiwara, H., Murai, T., Nasrun, M.W., Wiguna, T., Bardosono, S., Sekartini, R., Sarasvita, R., Murtani, B.J., Sen, L.T., Firdaus, K.K., 2022. Altered Resting-State Network in Adolescents with Problematic Internet Use. J. Clin. Med. 11 (19), 5838 https://doi.org/10.3390/jcm11195838.

Tossell, C.C., Kortum, P., Shepard, C., Rahmati, A., Zhong, L., 2015. Exploring smartphone addiction: insights from long-term telemetric behavioral measures. Int. J. Interact. Mob. Technol. 9 (2), 37. https://doi.org/10.3991/ijim.v9i2.4300.

Ünsalver, B., & Aktepe, H. (2017). Internet addiction and impulsivity among university students. European Psychiatry, 41(S1), S397-S397. doi:10.1016/j.eurpsy.2017.02.458.

US Department of Health and Human Services. (2016). THE NEUROBIOLOGY OF SUBSTANCE USE, MISUSE, AND ADDICTION. Facing Addiction in America - NCBI Bookshelf. https://www.ncbi.nlm.nih.gov/books/NBK424849/.

Van Rooij, A.J., Prause, N., 2014. A critical review of "Internet addiction" criteria with suggestions for the future. J. Behav. Addict. 3 (4), 203–213. https://doi.org/10.1556/jba.3.2014.4.1.

Vandenbroucke, J.P., von Elm, E., Altman, D.G., Gøtzsche, P.C., Mulrow, C.D., Pocock, S.J., Poole, C., Schlesselman, J.J., Egger, M., STROBE Initiative, 2007. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): explanation and elaboration. *Epidemiol.* (Camb., Mass.) 18 (6), 805–835. https://doi.org/10.1097/EDE.0b013e3181577511.

Volkow, N.D., Wang, G.J., Tomasi, D., Baler, R.D., 2013. Obesity and addiction: neurobiological overlaps. Obes. Rev.: Off. J. Int. *Assoc. Study Obes.* 14 (1), 2–18. https://doi.org/10.1111/j.1467-789X.2012.01031.x.

Wang, L., Shen, H., Lei, Y., Zeng, L.L., Cao, F., Su, L., Yang, Z., Yao, S., Hu, D., 2017. Altered default mode, fronto-parietal and salience networks in adolescents with Internet addiction. Addict. Behav. 70, 1–6. https://doi.org/10.1016/j.addbeh.2017.01.021.

Wang, Y., Qin, Y., Li, H., Yao, D., Sun, B., Li, Z., Li, X., Dai, Y., Wen, C., Zhang, L., Zhang, C., Zhu, T., Luo, C., 2019. Abnormal functional connectivity in cognitive control network, default mode network, and visual attention network in internet addiction: a resting-state fMRI study. Front. Neurol. 10, 1006. https://doi.org/10.3389/fneur.2019.01006.

Wee, C.Y., Zhao, Z., Yap, P.T., Wu, G., Shi, F., Price, T., Du, Y., Xu, J., Zhou, Y., Shen, D., 2014. Disrupted brain functional network in internet addiction disorder: a resting-state functional magnetic resonance imaging study. PloS One 9 (9), e107306. https://doi.org/10.1371/journal.pone.0107306.

Wittek, C.T., Finserås, T.R., Pallesen, S., Mentzoni, R.A., Hanss, D., Griffiths, M.D., Molde, H., 2016. Prevalence and predictors of video game addiction: a study based

on a national representative sample of gamers. Int. J. Ment. Health Addict. 14 (5), 672–686. https://doi.org/10.1007/s11469-015-9592-8.
Wu, A.M., Cheung, V.I., Ku, L., Hung, E.P., 2013. Psychological risk factors of addiction to social networking sites among Chinese smartphone users. J. Behav. Addict. 2 (3), 160–166. https://doi.org/10.1556/JBA.2.2013.006.
Yoon, H., Kim, S., Ahn, H.S., Kim, S.H., 2015. Altered neural activity in the anterior and posterior insula in individuals with problematic internet use. Eur. Addict. Res. 21 (6), 307–314. https://doi.org/10.1159/000377627.
Younes, F., Halawi, G., Jabbour, H., El Osta, N., Karam, L., Hajj, A., Rabbaa Khabbaz, L., 2016. Internet addiction and relationships with insomnia, anxiety, depression, stress and self-esteem in university students: a cross-sectional designed study. PloS One 11 (9), e0161126. https://doi.org/10.1371/journal.pone.0161126.
Yuan, P., Raz, N., 2014. Prefrontal cortex and executive functions in healthy adults: a meta-analysis of structural neuroimaging studies. Neurosci. Biobehav. Rev. 42, 180–192. https://doi.org/10.1016/j.neubiorev.2014.02.005.